



P7: Analyzing Thematic Alignment in Scientific Journals AY 2025/2026

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Research Question

Scientific journals define their thematic boundaries through Aim & Scope section.

Is it possible to assess quantitatively the coherence between published articles and the declared scope?

This leads to the need of having a systematic and reproducible method to assess this evaluation.

Goal

- *Introduction of a quantitative framework which measures the semantic alignment between published articles and the declared scope.*
- *Detection of thematic drift over time and potential outliers.*

1.530

Articles

Jan 2021– Dec 2025

Range of time

Crossref API

Data source

Dataset attributes

title

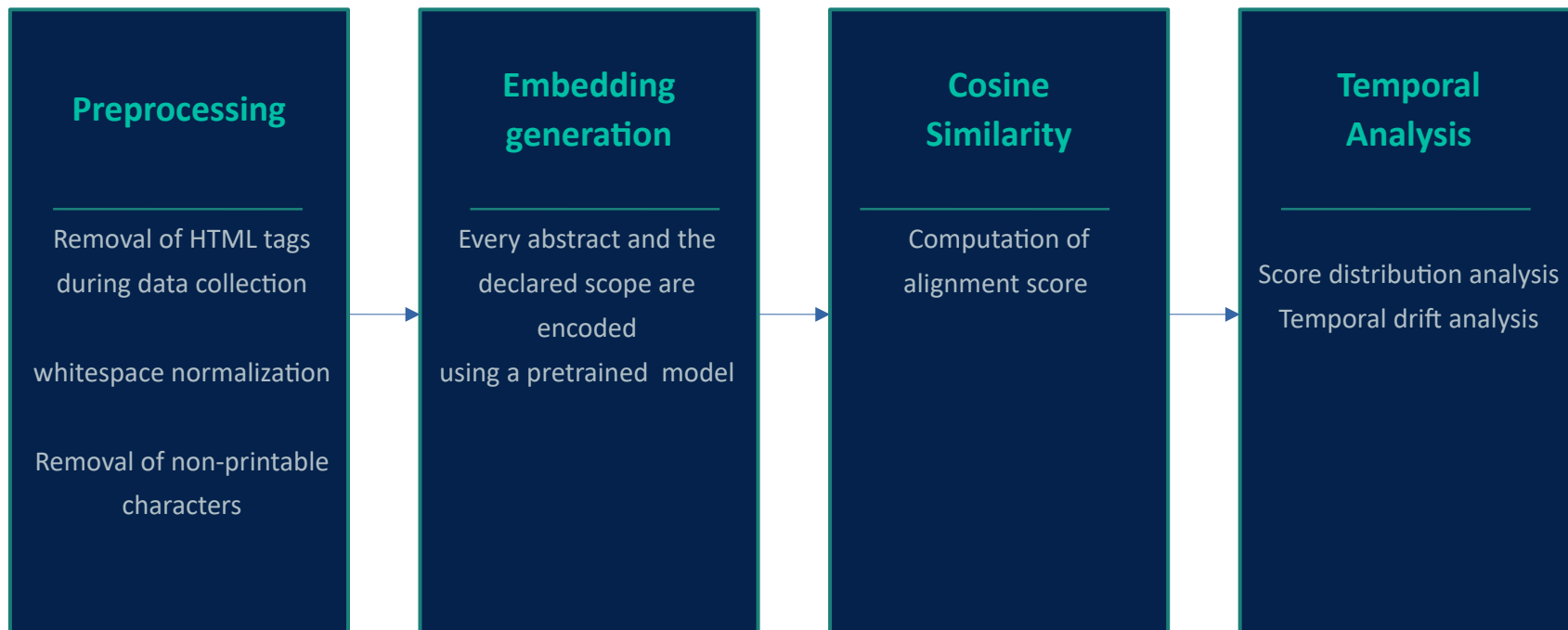
abstract

doi

year

Articles distribution per year

Year	Articles
2021	192
2022	276
2023	216
2024	292
2025	554





Model used

Allenai-specter, a transformer-based model specialized in scientific texts. It was fine-tuned using citation links between papers as a supervision signal.

Output: 768-dimensional embedding vectors.

Cosine similarity

Cosine similarity is computed between each article's abstract embedding and the journal's scope embedding.

Input:

1530 abstracts embeddings

1 scope embedding

Results – alignment score distribution



0.727

Mean

0.065

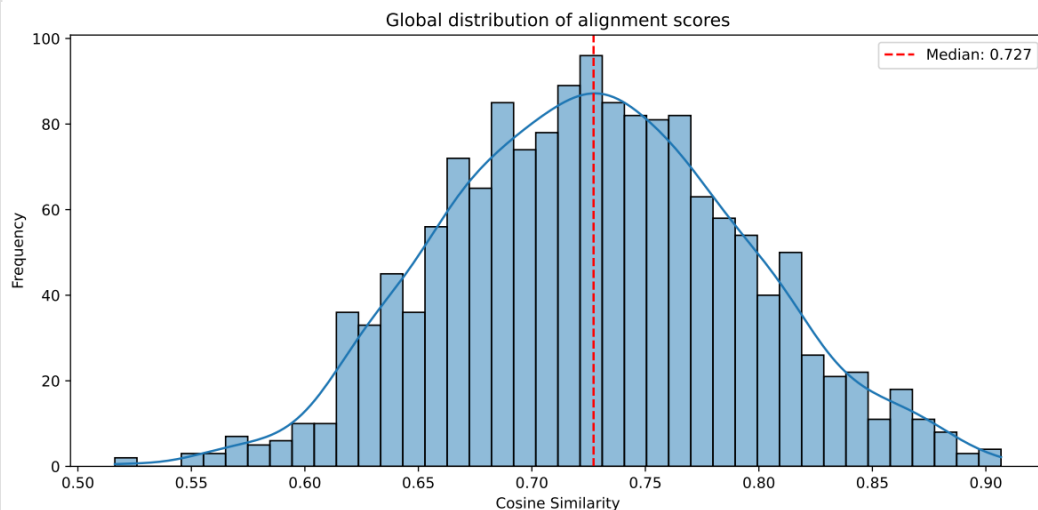
Std. Deviation

0.516

Min value

0.907

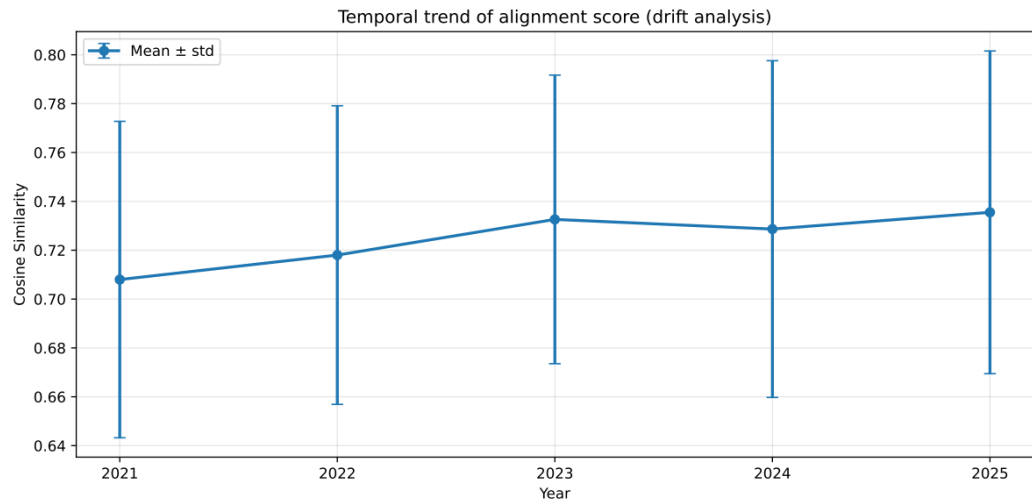
Max value



Interpretation

- The distribution is strongly concentrated around the mean with limited dispersion,
- The majority of published articles is semantically aligned to the declared scope.
- Aligned score < 0.6 potentially identifies articles deviating of scope(outliers),
- Aligned score > 0.85 indicates a strong alignment to the declared scope.

Results – Temporal analysis



Mean trend

Gradual increase from 0.708 (2021) to 0.736 (2025). No negative drifts, editorial coherence doesn't decrease over time.

Dispersion

Dispersion remains relatively stable over time, indicating limited thematic heterogeneity across publication years.



Quantitative framework

A reproducible method to evaluate semantic alignment between published articles and the declared scope of a selected scientific journal, based on quantitative semantic measures.

Possible improvements

A comparison of alignment score between multiple journals of the same sector.

Frontiers in Artificial Intelligence

A global mean of 0.727 indicates that the journal maintains editorial coherence during the selected period of time 2021-2025.

Limitations

The analysis is limited to 5 years and a single journal.